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NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)
Sixth Semester B.Tech (CBCS) Degree Examinations

May 2025

ME3002-1 – HEAT TRANSFER
(For ME)

Duration: 3 Hours

Max. Marks: 100

Note:

Part – A: Multiple Choice Questions: Answer all **Twenty** questions in the **OMR Sheet** provided. Each question carries equal marks.

Part – B: Descriptive Answer Questions: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II each** and **One full** question from **Unit – III**.

PART - A: MULTIPLE CHOICE QUESTIONS**20 Marks**

1. **Conduction Heat transfer takes place as per**
A) Fourier law
B) Zeroth law of Thermodynamics
C) First law of Thermodynamics
D) Second law of Thermodynamics
2. **Thermal conductivity of solid metals with rise in temperature**
A) decreases
B) increases
C) remains same
D) unpredictable
3. **Fin will serve the intended purpose if its effectiveness is**
A) Equal to zero
B) Less than unity
C) Equal to unity
D) More than unity
4. **If k is the thermal conductivity, ρ is the mass density and c is the specific heat then the thermal diffusivity of substance is given by**
A) $\rho c/k$
B) $k/\rho c$
C) $k c/\rho$
D) $k \rho/c$
5. **Which of the following dimensionless number gives an indication of the ratio of internal (conduction) resistance to the surface (convective) resistance?**
A) Biot number
B) Fourier number
C) Stanton number
D) Reynolds number
6. **Lumped parameter analysis for transient heat conduction is essentially valid for**
A) $Bi < 0.1$
B) $1 < Bi < 10$
C) $0.1 < Bi < 0.5$
D) It tends to infinity
7. **In the non-dimensional Biot number, the characteristics length is the ratio of**
A) Perimeter to surface area of solid
B) Surface area to perimeter of solid
C) Surface area to volume of solid
D) Volume of solid to its surface area
8. **According to lumped system analysis, solid possesses thermal conductivity that is**
A) Infinitely large
B) Infinitely small
C) Moderate
D) 50% small
9. **A body reflects entire radiation incident on it.**
A) Transparent
B) Black
C) Grey
D) White
10. **What is the dimension of thermal conductivity?**
A) $MLT^{-3}\theta^{-1}$
B) MLT^{-3}
C) $MLT^{-2}\theta^{-1}$
D) $MLT^{-3}\theta^{-2}$

11. Mark the system where heat transfer is given by forced convection.
- A) Fluid passing through the tubes of a condenser and other heat exchange equipment
B) Chilling effect of cold wind on warm body
C) Heat flow from a hot pavement to surrounding atmosphere
D) Heat exchange on the outside of cold and warm pipes
12. fluid is flowing over a hot plate of length L, the average heat transfer coefficient is given by
- A) $h_a = \int_0^L h_x \cdot dx$
B) $h_a = \frac{1}{L} \frac{d}{dx} (h_x)$
C) $h_a = \frac{1}{L} \int_0^L h_x dx$
D) None of the above
13. If there are 6 physical quantities and 3 fundamental units, then the number of pi terms are
- A) 3
B) 2
C) 1
D) 4
14. Free convection heat flow does not depend on
- A) Velocity
B) Density
C) Gravitational force
D) Coefficient of viscosity
15. In free convection heat transfer, Nusselt number is a function of
- A) Grashoff number and Prandtl number
B) Grashoff number and Reynold number
C) Reynold number and Prandtl number
D) Grashoff number, Prandtl number and Reynold number
16. Which dimensionless number has a significant role in forced convection?
- A) Reynolds number
B) Peclet number
C) Mach number
D) Prandtl number
17. The normal automobile radiator is a heat exchanger of the type
- A) direct contact
B) parallel flow
C) counter flow
D) cross flow
18. The fouling factor is
- A) A dimensionless number.
B) Used to take into account the resistance formed by the deposit.
C) Used in the flow of Newtonian fluid.
D) None of the above.
19. The heat transfer rate (Q) in a heat exchanger is given by
- A) $C_h (T_{hi} - T_{ci})$
B) $C_c (T_{hi} - T_{ci})$
C) $C_h (T_{hi} - T_{ho})$
D) $C_c (T_{hi} - T_{ho})$
20. Which of the following statements are true regarding heat exchanger
- A) The performance of heat exchanger depends upon its size.
B) Heat transfer in a heat exchanger usually involves convection in each fluids and conduction through the wall separating the two fluids.
C) Heat exchanger obeys both first and second law of thermodynamics.
D) All the above

PART - B: DESCRIPTIVE ANSWER QUESTIONS

Unit - I		Marks	BT*	CO*	PO*
1. a)	Derive an expression for heat flow and thermal resistance of the composite wall.	8	L*1	1	1
b)	A cylindrical stainless-steel ingot, 15 cm in diameter and 40 cm long passes through a heat treatment furnace which is 6 m in length. Temperature of furnace gas is 1250°C. Initial ingot temperature is 100°C. Combined radiant and convective H.T. coefficient is 100 W/m²K. Calculate the maximum speed at which the ingot moves through the furnace if it must attain 850°C temperatures. Take thermal conductivity of stainless steel is 45 W/mK and thermal diffusivity, $\alpha = 0.46 \times 10^{-3} \text{ m}^2/\text{s}$.	8	L1	2	1

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|-------|--|---|----|---|---|
| 2. a) | Derive an expression for one dimensional steady state heat flow in a hollow cylinder without heat generation. | 8 | L2 | 1 | 2 |
| b) | Derive the expression for temperature distribution, instantaneous heat transfer rate and overall heat transfer rate through a system with negligible internal resistance. | 8 | L2 | 2 | 1 |
| 3. a) | A wall of the furnace is made up of an inside layer of silica brick 120 mm thick covered with a layer of magnesite brick 240 mm thick. The temperature at the inside surface of the silica brick wall and the outside surface of the magnesite brick wall are 725 °C and 110 °C. The contact thermal resistance between the two walls at the interface is 0.0035 °C/W per unit wall area. If the thermal conductivities of silica and magnesite bricks are 1.7 W/m °C and 5.8 W/m °C. Calculate (a) The rate of heat loss per unit area of walls. (b) Temperature drop at the interface. | 8 | L3 | 1 | 1 |
| b) | An electrical cable 1.2cm outer diameter has to be insulated with rubber (K=0.15W/mK). The cable is to be located in air (h=8.7W/m²K) at 18 °C. Calculate: (a) The heat dissipation from the bare pipe. (b) Maximum heat dissipation. (c) The critical thickness of insulation assuming a cable surface temperature of 70 °C. | 8 | L3 | 2 | 2 |

Unit – II

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|-------|---|---|----|---|---|
| 4. a) | For a flat plate Explain velocity and thermal boundary layer with the help of a neat sketch. | 8 | L1 | 3 | 1 |
| b) | Air at atmospheric pressure and 30°C enters a 12m long 1.5cm diameter tube with a mass flow rate of 0.1kg/s. The tube surface is maintained at a uniform temperature of 90 °C. Calculate the average heat transfer co-efficient and rate of heat transfer to the air. | 8 | L3 | 4 | 2 |
| 5. a) | Using Buckingham's π theorem, obtain the relationship between various non-dimensional numbers for free convection heat transfer. | 8 | L1 | 4 | 1 |
| b) | An approximate expression for the temperature profile for laminar flow along a flat plate is given by $\theta(x, y) = \frac{3}{2} \left(\frac{y}{\delta(t)} \right) - \frac{1}{2} \left(\frac{y}{\delta(t)} \right)^3$ where $\theta(x, y) = \frac{T(x, y) - T_w}{T_\infty - T_w}$, T_w = temperature of the plate, T_∞ = temperature of the free stream fluid flowing over the plate, $T(x, y)$ = temperature at any location in the thermal boundary layer. $\delta(t)$ is thermal boundary layer thickness given by $\frac{\delta(t)}{x} = 4.53 \text{ Rex}^{-0.5} \text{ Pr}^{-1/3}$. Develop an expression for average Nusselt no. and average heat transfer coefficient. | 8 | L3 | 3 | 2 |
| 6. a) | Explain Grey bodies, white bodies and black bodies in terms of absorptivity, emissivity and transmittivity. | 8 | L1 | 3 | 1 |
| b) | A rectangular duct 30x20 cm in cross section carries cold air. The temperature of outer surface of duct is 5 °C and the surrounding air temperature is 25 °C. Find the heat gained by the duct assuming 1m length is exposed to air in vertical position. | 8 | L3 | 4 | 2 |

Unit – III

- | | | | | | |
|-------|---|---|----|---|---|
| 7. a) | With a neat sketch of boiling curve for water, explain the regimes of pool boiling. | 8 | L1 | 5 | 1 |
|-------|---|---|----|---|---|

- b) A square array of 400 tubes of 15 mm diameter is used to condense steam at atmospheric pressure. The tube walls are maintained at 88 °C by a coolant flowing through tubes. Calculate the amount of steam condensed per hour per unit length of the tubes.

8 L3 5 2

8. a) Derive an expression for Log mean Temperature Difference of counter flow heat exchanger.

8 L2 5 1

- b) A tubular heat exchanger consists of 200 tubes each 20 mm outer diameter & 5m length. Hot fluid flows inside the tube and cold over it and in the same direction of hot fluid. The overall heat transfer coefficient based on outside diameter is 320 J/s m² °K. Determine the outlet temperature of both fluids and total heat transfer using the data given below. $T_{hi} = 120^\circ\text{C}$, $\dot{m}_h = 20\text{ kg/s}$, $C_{ph} = 2000\text{ J/kgK}$, $T_{ci} = 20^\circ\text{C}$, $\dot{m}_c = 5\text{ kg/s}$, $C_{pc} = 4000\text{ J/kg K}$.

8 L3 5 2

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
